

EmoMoconvq: Parameter-Efficient Emotion Conditioning for Physics-Based Motion Generation

Final Project Report

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Abstract—We study whether MoConVQ [8], a frozen physics-based text-to-motion model, can be retrofitted with discrete emotion control via parameter-efficient fine-tuning. EmoMoconvq adds a 0.79M-parameter emotion branch to T2M-MoConGPT’s text features and distills supervision from baseline responses to emotion-bearing text. In the strongest five-class emotion-module-only variant, changing only the label increases mean pairwise token disagreement from 0.560 under baseline sampling noise to 0.762 (1.36 $\times$; paired $t(49) = 4.58$, $p = 3.2 \times 10^{-5}$; bootstrap 95% ratio CI [1.19, 1.59]). A separate focused three-class run obtains 39.3% token-classifier accuracy versus 33.3% chance, although the classifier’s held-out validation accuracy is only 26.7%, so we treat that result as exploratory. After frozen physics decoding, six affective kinematic features show no emotion effect (between-group ANOVA $p > 0.7$; repeated-measures ANOVA $p > 0.39$; paired tests $p > 0.26$). These results establish that the control changes discrete outputs but do not establish that those changes encode perceptible emotion. We present a frozen-decoder bottleneck as the leading mechanistic hypothesis, discuss alternative explanations, and release code, checkpoints, videos, and an interactive Gradio demo.

I. INTRODUCTION

Physics-based motion generators are increasingly competitive with kinematic-only models because they guarantee dynamic plausibility: the generated motion is executed by an articulated character under contact and gravity. The current state of the art, MoConVQ [8], generates physically simulated motion from text via a residual-VQ representation combined with a GPT-style transformer (T2M-MoConGPT) and a frozen physics tracking policy.

This expressive power comes with a control limitation: the model takes unstructured natural-language text as its only conditioning signal. A downstream system that wants to “set the emotional state” of an animated character must phrase that state in text. This is brittle (the exact wording matters; many emotion-action pairs are out-of-distribution) and it conflates emotion with all other semantic content the text carries.

Our question. Can we add a small *discrete* emotion interface to MoConVQ’s existing text-to-motion stack, such that the same text prompt can be rendered in different emotional styles by setting an explicit `emotion` flag?

Our approach. We freeze the entire pretrained MoConVQ model and attach a small emotion embedding module (~ 0.8 M trainable parameters) that perturbs the T5 text feature before it enters the temporal cross-attention layers (§III). Because no

public dataset of emotion-labelled physical motion exists in MoConVQ’s character format, we adopt a *distillation* strategy: we run the baseline T2M-MoConGPT on text that already contains explicit emotion adverbs (*walks happily*, *walks sadly*...) to produce token sequences, then train EmoMoconvq to reproduce those same token sequences from the *stripped* text plus the discrete emotion label (§III-C). Through three progressive variants we tighten the design until the emotion label is, by construction, the only input that distinguishes targets sharing a base caption (§III-D).

Our finding. Token-level evaluation (§IV-B) shows that the discrete label changes the model output on held-out HumanML3D [2] captions: in the emotion-module-only model, variation is 1.36 $\times$ baseline stochasticity and is significant in a caption-paired test. A separate three-class model obtains 39.3% exploratory classifier accuracy (33.3% chance). However, kinematic tests on physically decoded motions (§IV-C) find no emotion effect on six affective features. We therefore investigate a **physics-decoder bottleneck** (§V) as the primary explanation, while retaining two alternatives: the token differences may not be emotion-semantic, and the selected kinematic features may miss perceptually relevant changes.

Contributions. (i) A clean, non-invasive emotion-conditioning architecture for MoConVQ that is a strict parameter superset of the pretrained checkpoint. (ii) A reproducible three-stage distillation methodology that exposes the emotion module’s true capability without confounding text-memorisation. (iii) A token-level vs. motion-level decomposition that narrows where the control signal is lost and exposes the limits of token-only evaluation. (iv) A carefully bounded negative result with a testable decoder-bottleneck hypothesis and concrete follow-ups (decoder ablation and motion supervision). (v) A working Gradio interactive demo and 15-clip qualitative comparison video set.

II. RELATED WORK

Physics-based motion generation. MoConVQ [8] encodes motion into residual VQ tokens and decodes them via a model-based RL policy that drives an ODE physics simulator, trained on 23 hours of motion data. Its frozen encoder and codebook make it well-suited for lightweight downstream adaptation, which we exploit.

Text-to-motion generation. T2M-GPT [9] and MDM [7] generate kinematic motion from text using discrete transformers and diffusion respectively. Both are trained on HumanML3D [2], are not physics-based, and do not expose an explicit emotion control.

Style and emotion conditioning. ACTOR [5] conditions a transformer VAE on action labels. Aberman et al. [1] learn motion style transfer from an unpaired, style-labelled corpus, while 100STYLE [4] provides 100 locomotion styles for real-time modelling. These systems still depend on motion data whose skeleton and labels fit their training pipeline. Our approach instead uses text-prompt distillation and leaves the pretrained physics backbone unchanged.

Parameter-efficient adaptation. We draw on the LoRA/adaptor family [3] in spirit. Our emotion module is even lighter: $\sim 0.8\text{M}$ parameters with the rest of the model frozen. Path B of our experiments (§III-D) is essentially a $\sim 0.4\%$ adapter.

III. METHODOLOGY AND SYSTEM DESIGN

A. System overview

MoConVQ’s text-to-motion pipeline has three frozen components: (1) a 1D convolutional motion encoder + 8-layer residual VQ codebook ($N = 512$, $d = 768$); (2) T2M-MoConGPT, a dual transformer with a 12-layer temporal block and a 5-layer depth block predicting RVQ indices; and (3) a deconvolutional decoder + mixture-of-experts tracking policy + ODE physics simulator. We touch only the temporal transformer’s input (its T5 text feature) plus, optionally, its top few layers.

B. Emotion embedding module

Let $\mathcal{C} = \{\text{neutral}, \text{happy}, \text{sad}, \text{angry}, \text{fearful}\}$ be the 5-class emotion set. We learn an embedding table $E_e : \mathcal{C} \rightarrow \mathbb{R}^{d_e}$ ($d_e = 512$) and a two-layer MLP projection $\text{MLP}_{\text{proj}} : \mathbb{R}^{d_e} \rightarrow \mathbb{R}^{d_{\text{T5}}}$ that lifts the emotion vector to the 1024-dimensional T5-large [6] feature space consumed by cross-attention. The conditioning signal seen by every temporal block is

$$f_{\text{cond}} = f_{\text{T5}} + \text{MLP}_{\text{proj}}(E_e(c)). \quad (1)$$

The last linear of MLP_{proj} is zero-initialised so that $f_{\text{cond}} = f_{\text{T5}}$ at step 0 and the model degrades gracefully to the pretrained baseline. The whole module is implemented as a subclass of MoConVQ’s transformer; the upstream source code is unmodified. Loading the pretrained checkpoint into the subclass with `strict=False` leaves exactly the 5 emotion tensors uninitialised and produces no unexpected keys.

C. Distillation training objective

A complete public dataset of (text, emotion, MoConVQ-tokens) triples does not exist (§IV-A). We instead distil from the baseline:

- 1) Pick a set of *base* captions b that contain no emotion words.

- 2) For each base b and each emotion $e \in \mathcal{C}$, form the *augmented* caption $b^{(e)}$ by appending the canonical emotion adverb (*walks* $\rightarrow$ *walks happily*).
- 3) Run the frozen baseline T2M-MoConGPT on each $b^{(e)}$ to obtain a target token sequence $I^{(b,e)}$ and the corresponding quantised latent sequence.
- 4) Train EmoMoconvq to predict $I^{(b,e)}$ given (b, e) – crucially, the *stripped* base text plus the emotion label, never the augmented text.

The training objective is the same per-token NLL loss as MoConGPT, implemented from scratch (MoConVQ does not release training code).

Because b is the same across all five emotions but the targets $I^{(b,e)}$ differ, the text input alone cannot predict the target: the model is *forced* to use the discrete emotion label.

D. Three progressive experimental variants

We ran three increasingly tight variants to disentangle “the module learnt something” from “the module is the only thing that could have learnt it”.

v1 – uncontrolled distillation, top-4 layers. 463 unique emotion-bearing captions filtered from HumanML3D, one emotion per caption. The model can in principle memorise the text $\rightarrow$ tokens map purely from text features and ignore the emotion label. 47.3M trainable parameters (24.3% of the model).

v2 – 5-variant distillation, top-2 layers. 100 base captions $\times$ 5 emotions = 500 samples. Same base text, five different targets, so the emotion label is the only distinguishing input. Train/val split is *base-stratified* so held-out captions are never seen in training. 24.0M trainable (12.4%).

Path B – emotion-module only. v2 dataset, but every temporal layer is also frozen; only E_e and MLP_{proj} are trainable. 0.79M trainable parameters (0.41%).

3-class D’. v2 with the most challenging classes (*sad*, *angry*) removed – the proposal’s pre-registered fallback. 300 samples on {neutral, happy, fearful}; chance level 33.3% (vs. 20%).

IV. EXPERIMENTS AND EVALUATION

A. Datasets

Why distillation in the first place. The original proposal called for HumanML3D emotion-labelled motions plus CMU emotional-gait sequences. Two practical blockers emerged. (1) HumanML3D ships only text annotations; the motion files require AMASS registration and large download (40+ GB) plus a non-trivial SMPL $\rightarrow$ MoConVQ retargeting pipeline. (2) MoConVQ uses a custom 22-joint skeleton with non-standard joint names (`pelvis_lowerback`, `rTorso_Clavicle`, ...); its BVHToTargetBase loader does name-matched pose copying, not real retargeting. We verified this by attempting to load a public stylised-locomotion BVH from 100STYLE [4] (*Angry/Angry_BR.bvh*), which failed immediately with `KeyError: ‘pelvis_lowerback’`. A real

retargeter would require multi-week engineering, so we chose the distillation route, with full disclosure of this scope change.

Emotion-keyword corpus. We scan HumanML3D’s 23,384 training captions ($\approx 70,000$ caption lines) for any word from our hand-curated emotion lexicon (e.g. *happily*, *joyfully*, *sadly*, *angrily*, *fearfully*...). The resulting 463 caption–motion-id pairs form the v1 corpus. Class counts are imbalanced (happy 126, sad 28, angry 132, fearful 177); 229 of 463 are left/right mirror copies, so HumanML3D’s built-in mirror augmentation is already baked in.

Distillation v2 corpus. We sample 100 *base* captions from HumanML3D’s training pool that satisfy: no emotion keyword present, contains at least one motion-verb, length $\in [5, 25]$ words, motion id not in the v1 set. For each base and each of the 5 emotions, the baseline T2M-MoConGPT samples a target (I, V) pair at a fixed seed. Output: 500 (base_text, emotion, I, V) triples; sequence length distribution min 10, median 50, max 50, mean 40.4.

B. Token-level compositional evaluation

We test compositional generalisation: does the trained EmoMoconvq actually *use* the discrete emotion label on captions that were not in training? We sample 50 neutral test captions from the HumanML3D training pool that (a) contain no emotion keyword and (b) have a motion id not in any of our distillation sets.

For each test caption c , EmoMoconvq is run once per emotion at a fixed seed, giving 5 outputs that differ *only* in the discrete emotion label. The baseline T2M-MoConGPT is run 5 times on the same c with different seeds, giving 5 outputs that differ only by sampling noise. Two metrics:

Separation ratio = (mean pairwise token-disagreement among EmoMoconvq’s 5 outputs) $\div$ (mean pairwise disagreement among baseline’s 5 outputs). Values > 1.0 mean the emotion knob produces *more* variation than baseline stochasticity.

For inference, we retain one emotion and one baseline disagreement score per caption, compare them with a paired t -test, and bootstrap captions (50,000 resamples, fixed seed) to obtain a 95% confidence interval for the ratio. This paired design controls for caption-specific motion complexity.

Classifier accuracy. We train a small bi-GRU (tokens $\rightarrow$ emotion) on the distillation dataset and apply it to EmoMoconvq’s compositional outputs. Per-class accuracy says whether the model’s outputs resemble the intended class in that learned token space. Because this metric is meaningful only when the classifier generalises to held-out distillation samples, we report its validation accuracy and treat generation accuracy as exploratory when validation is at or below chance.

Figure 1 summarises both metrics across the four experimental variants.

The trends are consistent at the disagreement level: v1 (loose, large-capacity) is near baseline noise (ratio 0.96); v2 reaches 1.09; and limiting capacity to the emotion module itself (Path B) reaches 1.36. For Path B, mean disagreement

TABLE I
ONE-WAY ANOVA: KINEMATIC FEATURE $\sim$ EMOTION LABEL (20 CAPTIONS $\times$ 3 EMOTIONS). NO FEATURE REACHES THE CONVENTIONAL $p < 0.05$ THRESHOLD; EFFECT SIZES $\eta^2 \leq 0.011$.

Feature	F	p	η^2
vertical_energy	0.000	1.000	0.000
avg_height	0.036	0.965	0.001
avg_speed	0.178	0.838	0.006
step_freq	0.314	0.732	0.011
lean_angle	0.246	0.783	0.009
arm_swing	0.058	0.944	0.002

is 0.762 versus 0.560 for baseline sampling. The caption-paired difference is significant ($t(49) = 4.58$, $p = 3.2 \times 10^{-5}$; Wilcoxon $p = 3.9 \times 10^{-5}$), and the bootstrap 95% CI for the ratio is [1.19, 1.59]. The emotion-embedding table moves by L_2 norm 1.35 over training, about $8\times$ the v1 shift. Thus the label reliably changes token sequences, although disagreement alone does not show that the changed tokens express the intended emotion.

3-class focus. In the focused setup ($\{\text{neutral, happy, fearful}\}$), generation accuracy is 39.3% versus 33.3% chance; per-class values are 30% / 50% / 38% (Fig. 2). However, the same classifier obtains only 26.7% on its held-out distillation split, below chance. The 39.3% result is therefore exploratory rather than confirmatory. Moreover, this model’s token separation ratio is 1.10, with a bootstrap 95% CI [0.90, 1.35], so it does not independently replicate Path B’s strong separation result.

C. Motion-level kinematic ANOVA

The above shows the *tokens* differ. The next question is whether the differences survive physics decoding. We ran EmoMoconvq through MoConVQ’s frozen physics decoder on 20 compositional test captions for the 3 active emotions (60 simulated motions total) and computed six features from the published affective-motion literature: vertical energy (variance of root height; high $\Leftrightarrow$ bouncy), mean root height, mean horizontal speed, foot-strike frequency, mean trunk-lean angle, and arm-swing range.

Each feature was tested for a per-emotion effect with one-way ANOVA (across-caption) and with paired analysis (within-caption: subtracting the neutral baseline of the same caption). The result is unambiguous (Table I, Fig. 3):

Within-caption analysis tells the same story (Fig. 4). A repeated-measures one-way ANOVA gives $p \in [0.391, 0.823]$ and partial $\eta^2 \leq 0.048$ across the six features. Paired t -tests against neutral give $p \geq 0.268$, and difference signs are split 40–60% in either direction. Under these predefined features and this sample size, we find no systematic motion-level emotion effect.

D. Qualitative comparison & interactive demo

We render side-by-side comparison videos for 5 test captions $\times$ 3 emotions = 15 clips, decoded through MoConVQ’s physics simulator, visualised

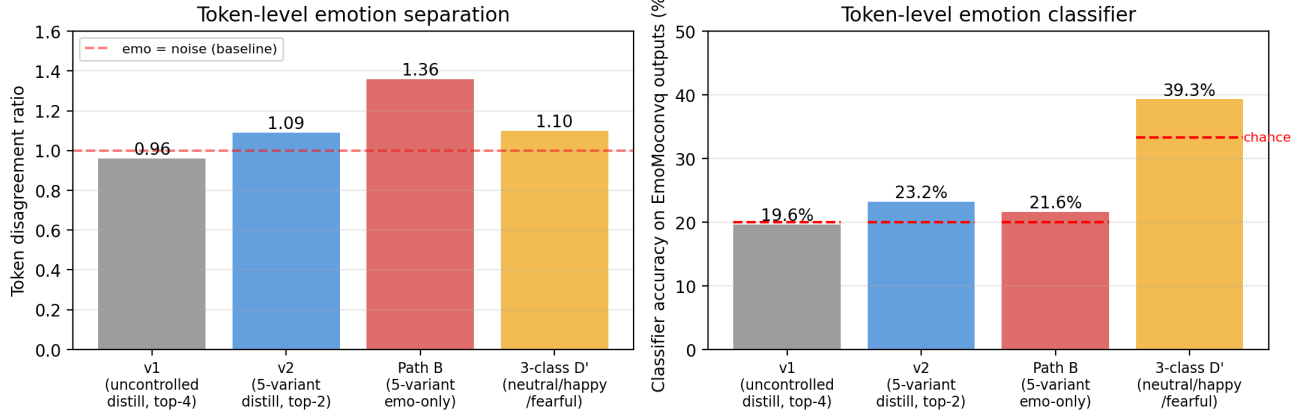


Fig. 1. Token-level emotion separation across the four progressive variants. Left: pairwise token disagreement caused by changing the emotion label, normalised by baseline stochasticity. Right: descriptive classifier accuracy on compositional-test outputs versus chance. The classifier panel must be read with its held-out validation accuracy, discussed in the text.

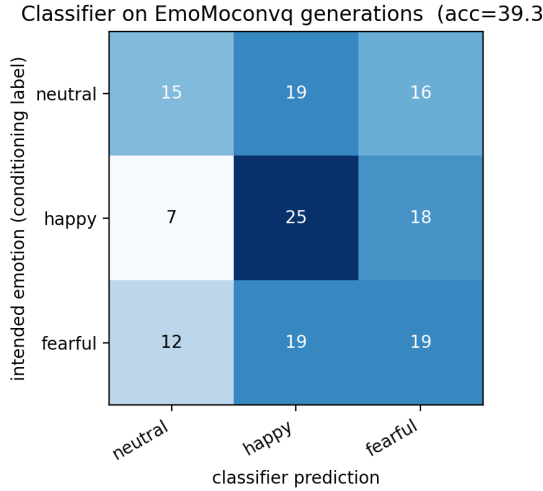


Fig. 2. Confusion matrix of the 3-class token classifier on compositional outputs. Accuracy is 39.3% (chance 33.3%), but held-out classifier validation is 26.7%; consequently this matrix is descriptive, not evidence of reliable emotion recognition.

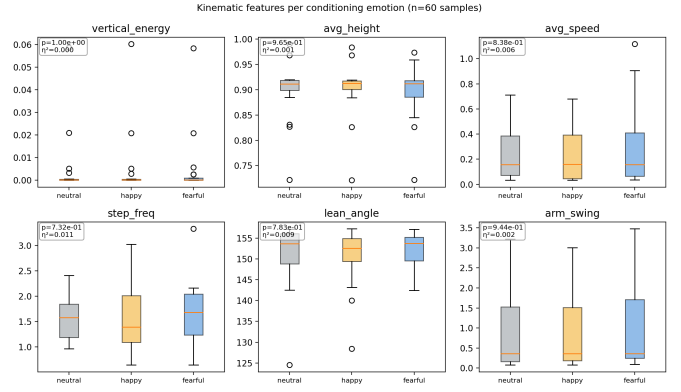


Fig. 3. Per-emotion distributions of the six affective kinematic features over 20 compositional test captions. Boxes are essentially indistinguishable across emotions; the embedded p -values are all > 0.7 .

in Blender with a per-bone tracked skin geometry (`render_qualitative_videos.py`). Some clips exhibit visible trajectory or timing differences, but the differences are inconsistent across captions and do not reliably map to the requested emotion. We did not run a sufficiently powered, blinded user study; the videos are therefore qualitative examples rather than perceptual evidence. They are released alongside the code for independent inspection.

We also ship a Gradio web demo (`Script/gradio_demo.py`) that accepts a free-form text prompt, an emotion label and a seed, and returns a downloadable BVH file produced by EmoMoconvq through the physics simulator. The first request loads the agent,

T5-large, and our checkpoint (~ 2 min); subsequent requests take ~ 10 s each. This is a real, interactive, physics-based emotion-conditioned generator.

V. RESULTS AND DISCUSSION

A. Interpreting the token-motion gap

The pattern in our results is:

- EmoMoconvq’s emotion module *is* trained: emotion-embedding L_2 -shift is large; the module receives non-trivial gradients.
- Its *token* outputs differ between conditioning labels and the Path B difference exceeds baseline noise (1.36 $\times$ with a confidence interval excluding 1).
- The independent token classifier is not reliable enough to establish that these differences encode emotion semantics.
- Yet the *decoded physical motion* is statistically indistinguishable across emotions on six standard affective features.

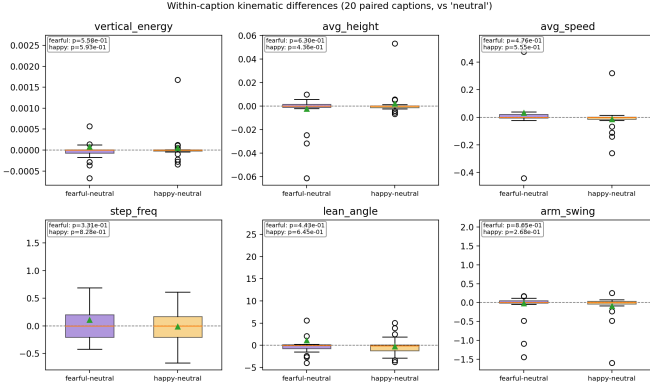


Fig. 4. Within-caption paired differences (vs. the same caption’s neutral baseline). Bars centred on zero; sign-consistency at $\approx 50\%$ indicates no systematic emotion effect.

Our leading hypothesis is that the frozen **physics-tracking policy** acts as a regression toward its training distribution. MoConVQ’s tracker was trained, via model-based RL, to follow latent trajectories while maintaining physical plausibility. A new conditioning branch can change upstream tokens without providing the tracker motion-level supervision for the corresponding style. The tracker may therefore map distinct upstream sequences to similar joint torques, contacts, and centre-of-mass trajectories. We call this candidate mechanism the *decoder bottleneck*.

The current experiments do not causally localise the loss to the tracker. Two alternatives remain. First, token disagreement may lie in code dimensions that are redundant or unrelated to emotion before physics decoding. Second, our six summary features may miss local timing, gesture, or trajectory cues that a human observer could perceive. Distinguishing these explanations requires an ablation that compares token, ConvVQ-latent, pre-simulation target motion, and final simulated motion, plus a blinded perceptual study. Thus the decoder bottleneck is a testable diagnosis, not a proven architectural law.

B. Why the proposal’s planned 60% target was not reached

The proposal targeted FID degradation ≤ 0.5 , emotion-classifier accuracy $\geq 60\%$, and $\geq 70\%$ user-study recognition. With the observed token–motion gap, these targets were not reached. The 39.3% three-class generation score is exploratory because classifier validation is below chance, and the motion-level tests are null. We report these outcomes instead of presenting the descriptive score as successful emotion control.

C. What this means for future work

The decoder-bottleneck hypothesis motivates three concrete follow-ups. Each requires substantial extra engineering that was beyond the scope of this course project.

(i) **Decoder ablation and tracker adaptation.** First measure where separation disappears by comparing ConvVQ latents, target poses, and simulated poses. If the tracker is responsible, adapt it with policy-gradient updates or a

differentiable-physics surrogate while jointly training the emotion branch. MoConVQ’s released code does not expose this training path, so enabling it is a non-trivial infrastructure project.

(ii) **Emotion-preservation loss in the tracking objective.** Adding a term that penalises kinematic divergence from an emotion-classifier’s prediction. This requires a real, motion-data emotion classifier, which in turn requires real emotion-labelled motion data – the same Path C blocker we hit at evaluation time.

(iii) **End-to-end training on emotion-labelled physical motion.** The proposal’s original ambition. The barrier is the MoConVQ-character retargeting problem we documented in §IV-A. Building a real retargeter (joint-topology mapping, IK foot-contact correction, scale normalisation) for an external emotion-mocap dataset is a multi-week engineering investment by itself; we judged it out of scope for the remaining time budget after fixing the data pipeline.

VI. CONCLUSION

We built a clean, minimal, parameter-efficient emotion-conditioning extension to a state-of-the-art physics-based text-to-motion model, and tested it under progressively tighter experimental controls. The emotion module verifiably changes token outputs beyond baseline sampling noise in the strongest controlled variant. However, the classifier evidence for emotion semantics is inconclusive, and the decoded motions show no significant variation on six affective kinematic features. A frozen tracking decoder is the leading explanation for this gap, but causal confirmation requires intermediate-latent ablations and perceptual evaluation. The project therefore contributes both a working conditioning system and a reproducible, bounded negative result that identifies the next experiments needed for real emotion-expressive physical motion.

RELEASED ARTEFACTS

Code, three trained EmoMoconvq checkpoints (v1, v2, 3-class), v1 & v2 distillation datasets, 15 qualitative comparison videos, and the Gradio demo are at <https://github.com/Sunkw1224/EmoMoconvq>.

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